Mega Plus PCB, Inc.
The Flexible Circuit & Rigid PCB Specialists

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LED and Lighting PCB

An LED & lighting PCB is a specialized printed circuit board engineered to power, control, and thermally manage LED components and lighting systems. These PCBs ensure stable electrical performance while meeting the demanding efficiency and reliability requirements of modern lighting applications. Effective heat dissipation is critical in LED systems. To address this, LED PCBs, commonly built using aluminum substrates, act as efficient heat sinks, improving thermal management, electrical stability, and the overall lifespan of lighting products. As a trusted LED PCB manufacturer, Mega Plus PCB delivers high-quality LED circuit boards for a wide range of applications, including automotive, industrial, commercial, and residential lighting. We support everything from rapid prototyping to large-scale production. With our expertise in LED lighting PCB fabrication, we provide solutions tailored to specific performance requirements, ensuring reliability, consistency, and manufacturability at

every stage of production.

Request a Quote

Our Lighting (LED) PCB Fabrication Capabilities

Getting specifications right from the start is what makes a lighting (LED) PCB fabrication project run smoothly. The table below outlines Mega Plus PCB's manufacturing capabilities, covering the full scope of our LED PCB Services.

Specification	Capability	Description
PCB Types		
Single-Sided, Double-Sided, Multi-Layer, Metal Core		
We manufacture all major LED PCB types, with metal core boards suited for high-power, thermal-critical applications.		
Board Size		
Up to 1105 x 610mm (44" x 24")		
Our fabrication supports LED PCBs up to 1105x610mm. Custom sizing is available for specific project needs.		
Board Thickness		
0.1mm to 8mm (4 to 320 mil)		

We cover single-sided boards from 0.1mm to 6.5mm and multi-layered boards from 0.2mm to 8mm.
Solder mask
Green, Black, White, Blue, Yellow, Red, Purple, Brown, Orange
Our solder mask options protect LED PCBs from environmental exposure while allowing application-specific color selection.
Drilling Techniques
Laser Micro Via, Controlled-Depth Drilling
We apply precision drilling methods enabling high-density routing for complex, tight-tolerance LED PCB layouts.
Surface Finishes
ENIG, Immersion Silver, ENEPIG
Our finishes deliver strong solderability, oxidation resistance, and long-term reliability for demanding LED systems.
Quality Certifications
ISO 9001, IPC Class 2/3, RoHS, ITAR, UL
We adhere to globally recognized standards at every stage of our lighting (LED) PCB fabrication process.
Silkscreen
White, Yellow, Black, Blue, Red

Our silkscreen layer marks component placements and assembly references, supporting accurate LED PCB builds.
Copper Weight
Up to 17 oz (Inner and Outer Layers)
We support up to 17 oz copper on inner and outer layers, addressing high current-load LED lighting demands.

Step-by-Step LED and Lighting PCB Manufacturing Process

At Mega Plus PCB, we follow a precise and efficient lighting (LED) PCB fabrication process, ensuring that each LED PCB meets the required performance standards and quality benchmarks. Below is the detailed step-by-step process:

1

Design Review and Thermal Analysis

Our engineering team reviews the PCB design in detail, evaluating thermal flow, power distribution, and manufacturability to ensure the LED PCB performs efficiently under real operating conditions without overheating or performance loss.

2

Material Selection and PCB Stackup Planning

Based on the design review and application requirements, we select the most appropriate substrate materials including aluminum core, FR4, or high-frequency materials and define the PCB stackup. For high-power LED applications, thermal conductivity ratings of the dielectric layer are carefully matched to the LED junction temperature requirements and target thermal resistance specifications.

3

Precision PCB Fabrication

We execute fabrication using controlled drilling, precise plating, copper etching, lamination, and layer bonding processes, ensuring every LED PCB layer aligns accurately for stable electrical connectivity and long-term operational reliability.

4

Surface Finish and Soldermask Application

For most LED PCB applications, Electroless Nickel Immersion Gold (ENIG) is preferred for its excellent solderability, flat surface, and long shelf life. We also offer Organic Solderability Preservative (OSP) and Immersion Tin for cost-sensitive or specific process requirements. We apply a high-quality soldermask to protect copper traces and define pad openings for SMT LED components.

5

Inspection and Testing

We conduct thorough quality checks including AOI inspection, electrical validation, thermal performance testing, and reliability assessments, ensuring every LED PCB meets strict quality standards before final delivery and application use.



Beneficial Features of Our LED and Lighting PCBs

At MegaPlus PCB, our LED and lighting PCBs are designed to offer superior performance, reliability, and efficiency, making them the optimal choice for various lighting applications. Here are some key benefits that set our solutions apart from the competition.

- **Superior Thermal Management:** Our LED PCBs feature enhanced thermal control, efficiently dissipating heat to prevent damage and maintain consistent brightness, even in high-power applications.

- **High-Density Interconnects:** We support advanced interconnect designs with precise drilling and surface finishes, enabling complex layouts while maintaining signal integrity in compact lighting systems.
- **Long-Lasting Durability:** Our LED PCBs are built to withstand challenging environments, providing reliable, long-term operation in both indoor and outdoor settings, ensuring minimal downtime.
- **Customizable Designs:** We offer a wide range of design options, including custom shapes and sizes, to fit unique project needs, ensuring optimal integration into lighting systems.
- **Energy Efficiency:** Our LED PCBs maximize energy use while maintaining high output, contributing to energy savings and environmentally-friendly lighting solutions for both commercial and residential applications.

Applications of LED and Lighting PCBs

LED PCBs are integral in various lighting applications. At MegaPlus PCB, we specialize in producing Lighting (LED) PCBs that address the unique challenges of each sector. Our LED PCBs are engineered to deliver high performance in diverse environments, ensuring reliability, energy efficiency, and durability.

- **Residential and Commercial LED Lighting:** Our LED PCBs are used in bulbs, panels, and ceiling systems. They deliver stable illumination, reduced heat buildup, improved efficiency, and long operational life across residential and commercial spaces.
- **Automotive LED Lighting:** We produce LED PCBs for headlights, tail lights, and interiors. They ensure consistent brightness, vibration resistance, thermal stability, and dependable performance under continuous automotive operating conditions and stress.
- **Industrial and Outdoor Lighting:** Our LED PCBs support streetlights, industrial lamps, and high-power systems. They offer strong thermal endurance, moisture resistance, and reliable operation in demanding outdoor and industrial environments with continuous usage.

- **Smart Lighting Systems:** We manufacture LED PCBs for IoT-based lighting solutions. They enable programmable control, remote access, adaptive brightness, and energy optimization across smart homes, offices, and automated building systems.
- **Display and Decorative Lighting:** Our compact LED PCBs are used in display panels and decorative lighting. They provide design flexibility, uniform illumination, easy integration, and reliable performance in architectural and visual applications.



FAQs

▼ What is the functional difference between blind vias and buried vias in multi-layer LED PCB designs?

Blind vias connect outer to inner layers. Buried vias link inner layers only. Both reduce surface routing congestion, which is especially useful in high-density LED PCB layouts.

► How does controlled-depth drilling differ from standard through-hole drilling in LED PCB production?

► Can Mega Plus PCB fabricate LED PCBs with n layers?

► How does ENIG compare to ENEPIG for LED PC reliability?

► Is ITAR compliance relevant for LED PCB procu lighting projects?

MENU

HOME

PRODUCTS

FLEX PCB MANUFACTURER

2-LAYER FLEX PCB MANUFACTURER

4-LAYER FLEX PCB MANUFACTURER

6-LAYER FLEX PCB MANUFACTURER

LONG FLEX PCB

RIGID FLEX PCB

4-LAYER RIGID-FLEX PCB

6-LAYER RIGID-FLEX PCB

8 LAYER RIGID FLEX

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